

Task 2. Classes

Objective: during the first task you discovered data, how do they look like, what are possible relationships, etc. Now we are going forward to our main goal. The goal of this task is to discover interpretable market classes / regimes that separate Binance trades by their expected maker markout and turnover. A class is a binary or bucketed condition computed for every Binance trade using only information available at or before the trade timestamp. To put it as simply as possible we need features that will discriminate markout (30s, 120s, 300s, avg_markout) and turnover. Examples:

- trade happened within 5 seconds after a large buy liquidation
- trade happened when spread is in the top 5% of historical spread values
- trade happened after a liquidation cascade and in the same direction as liquidation pressure
- trade happened when distance to midprice is large and recent signed trade flow is one-sided

The purpose is not yet to build the final ML model. The purpose is to find reusable, statistically meaningful classes that can later be used for:

- manual filtering rules
- ML features
- debugging and interpreting strategy results
- selecting subsets of trades where the signal is especially strong

A good class should satisfy three properties:

- It has a clearly different distribution of markout compared to the unconditional population
- It has enough turnover to matter
- It is stable across symbols, days, and horizons, not driven by one outlier event

Data: <https://drive.google.com/file/d/1FpLBVKViVhvrFXw1vgBL743Y1IFWDwFD/view?pli=1>

From 2025-11-01 to 2026-04-28:

<https://drive.google.com/file/d/1XmxRsElei-vE8Gc5tkKs2wH4FJVRTevS/view>

Deliverables: Jupyter Notebook

- Clean, documented code with markdown explanations
- Executive summary with actionable insights and limitations of the analysis & data caveats

Assessment:

- Technical execution: data handling, code quality, visualization, etc.
- Analytical depth: casual relationship, pattern discovery, hypothesis generation, critical reflection, no data leakage etc.
- Communication & Deliverables: oral presentation, notebook structure, executive summary, reproducibility

Task 2.1

The list below is not exhaustive. Additional ideas are encouraged.

- Think about variables that may result in vivid change of distribution of markout and turnover
 - For example, the previous time you analyzed liquidations
 - Change of funding rate
 - Big orders
 - ...
- You need to consider different dimensions to construct distributions
 - For example, distribution of markout relative to distance to midprice
 - Bid-ask spread
 - Time of day
 - ...
- Overall, a good example of a class is “How did the distribution of markout by distance to midprice change before and after big liquidation.”